Linux File Management Commands

A practical guide covering essential Linux commands for directory and file management.

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1. `mkdir` (Make Directory)

- **Definition:** The `mkdir` command creates new directories (folders) in the file system.
- **Usage / Formula:**

```
mkdir [OPTIONS] DIRECTORY_NAME...
```

Examples:

- **Create a single directory:**

```
mkdir projects
```

- **Create multiple directories at once:**

```
mkdir docs images downloads
```

- **Create parent and child directories together (nested):**

```
mkdir -p website/assets/images
```

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2. `touch`

- **Definition:** The `touch` command creates empty files or updates the access and modification timestamps of existing files.
- **Usage / Formula:**

```
touch [OPTIONS] FILENAME...
```

Examples:

- **Create a new empty file:**

```
touch index.html
```

- **Create multiple empty files at once:**

```
touch style.css script.js README.md
```

- **Update timestamps of an existing file without changing content:**

```
touch notes.txt
```

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3. `rm` (Remove)

- **Definition:** The `rm` command deletes files and directories from the file system. *(Note: Deleted items bypass the trash bin and cannot easily be recovered).*
- **Usage / Formula:**

```
rm [OPTIONS] FILE_OR_DIRECTORY...
```

Examples:

- **Remove a single file:**

```
rm index.html
```

- **Remove a directory and all of its contents recursively:**

```
rm -r website/
```

- **Force removal of files without prompting for confirmation:**

```
rm -f old_file.txt
```

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4. `cp` (Copy)

- **Definition:** The `cp` command copies files or directories from one location to another.

- **Usage / Formula:**

```
cp [OPTIONS] SOURCE DESTINATION
```

Examples:

- **Copy a file in the same directory under a new name:**

```
cp index.html index_backup.html
```

- **Copy a file into a different directory:**

```
cp image1.jpeg website/assets/
```

- **Copy an entire directory and all its contents recursively:**

```
cp -r images/ assets/images_backup/
```

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5. `mv` (Move / Rename)

- **Definition:** The `mv` command moves files or directories to a different folder, or renames them if the destination is in the same path.
- **Usage / Formula:**

```
mv [OPTIONS] SOURCE DESTINATION
```

Examples:

- **Rename a file:**

```
mv image1.jpg image1.jpeg
```

- **Move a file into a directory:**

```
mv index.html website/
```

- **Move all files matching a pattern into another folder:**

```
mv lab6-challenge2/img*.* website/assets/
```